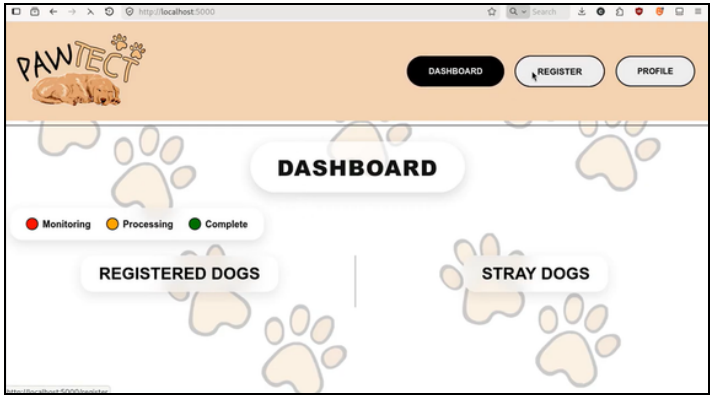
INTEGRAL PARTS

1. **PawTect Application**

   A program to **register local dogs**. Therefore, locate roaming registered dogs and stray dogs in a neighborhood, subdivision or barangay.

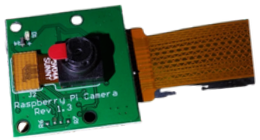


Access the program in the QR code on top

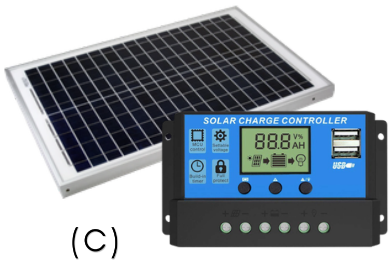
2. **Detections in the Form of a Bollard**

   A device that has a camera and a mini-computer for automatic detection of roaming dogs.

2A. Camera
2B. LoRa Atenna
2C. Solar System



(A)



(C)

(B)

USER MANUAL

PAWTECT

A COMMUNITY-BASED WIRELESS DOG DETECTION AND ALERT SYSTEM FOR STRAY AND MISSING DOGS USING COMPUTER VISION AND RASPBERRY PI

FOR THE BETTERMENT OF OUR DOG COMPANIONS

SET UP

1. PawTect Application

An execution file would be placed onto the online folder.

Dashboard

If detection occurs and dog is registered in the application, owners would be **automatically contacted**.

Defining the **legend**:

- Continuous Monitoring
- Identified, Sent a Notification
- Owner Replied
- Returns Back to Red After a Minute

Register

Input all the information of the owner and the dog. **Require the download of Telegram** for constant feedback.

Take a **clear picture** of their front, right, left, and back (bottom-side)

Make sure to **correctly input** the **contact number.**

Profile

Stores all registererd owner and dog's information in a page.

Long Range

Connect LoRa via **MicroUSB cable** directly to your main computing device.

2. Detections in the Form of a Bollard

A device would be given. There are hardware consideration during the placement.

Bollard

Deploy the bollards at strategic locations with **high stray dog activity** to maximize detection opportunities.

This may also be placed on **entrances and exits** of an area.

Camera

Position the bollard hole toward the designated detection field of view.

The camera captures 72.4°. Consider it in placement.

LoRa Atenna

Place the LoRa Antenna in an **elevated, unobstructed position** to maximize signal range and avoid interference.

The bottom surface of the Antenna is **magnetic**. This may be placed on poles at the street as seen at the side.

Solar System

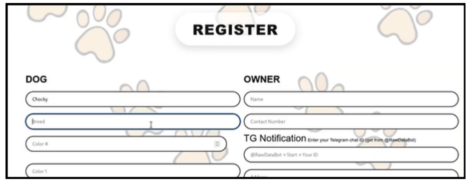
Position the solar panel in a sunlit, unshaded location for optimal charging efficiency.

Keep the charge controller protected from environmental elements inside the main housing.

PROCESS

1. Registration

Data will be stored after filling up the form. Images will be used for recognition process later.



2. Detection

This is not displayed in a monitor. It is running at the background as the power is on at the bollard device.

Identifies dogs from the environment and crops them. It will be sent via transmission line.

3. Recognition

The sent image would be compared to the database's images through feature extraction. This would compare the embeddings.

If similarity percentage is high from one of the database's profile, then it will contact the specific owner in SMS and Telegram.

If similarity percentage is low across the database, then it would be identified as a stray dog.

4. Contact

The owner would be contacted with an attached image of the photo detected for further inspections.

Reply with:

FOUND - This is my dog!
NOTMINE - This is not my dog!